

Architecture Design Studio Learning + Teaching Document

RMIT Architecture Design Studio Learning Framework

RMIT Architecture is known for ideas-led ventures design practice. Our mission and point of difference are founded on differentiated, curated and vertically integrated scholarship. Our ambition is to sustain contested areas of scholarly endeavours across the school. We believe that holding multiple, articulated positions leads to a productive scholarship environment. We understand these positions as points of intensity within a dynamic constellation structure rather than as fixed points.

Studio Title: Hong Kong Travelling Studio – Circular Tectonics

Studio Leaders: Dr Nic Bao + Harlan Guo

Sem/Year: Semester 2 2026

1. Studio Outline (250 words)

Studio Rationale

Circular Tectonics positions architecture within the urgent transition towards a circular built environment, where material reuse, digital technologies, and collaborative modes of construction are reshaping architectural practice. In response to increasing construction waste, resource depletion, and the environmental impacts of conventional building methods, the studio reconsiders tectonics through the lens of circular material systems and computational design.

Rather than viewing reclaimed materials as constraints, Circular Tectonics treats them as active generators of architectural form, structural logic, and construction strategy. Students will investigate how irregular material stocks can be digitally catalogued, analysed, and transformed through computational workflows integrating 3D scanning, parametric modelling, structural optimisation, digital fabrication, and mixed-reality-assisted assembly. Computation is understood not simply as a tool for formal expression, but as a means of negotiating relationships between material availability, fabrication constraints, structural performance, and architectural intention.

The studio also explores architecture as a collaborative and socially engaged practice. Through an international design-build partnership between RMIT University and The Chinese University of Hong Kong (CUHK), students will develop individual pavilion proposals and scaled physical prototypes before collectively fabricating and assembling a 1:1 installation using locally sourced reclaimed materials. By combining circular design principles with emerging fabrication technologies, the studio promotes an architectural approach in which environmental responsibility, material intelligence, and construction innovation become inseparable components of contemporary tectonic practice.

Studio Aims and Objectives

This studio investigates circular tectonic design methodologies that integrate reclaimed material systems, computational design, and digital fabrication within an iterative design-to-construction workflow. Its primary aim is to challenge conventional linear models of architectural production by positioning material availability, construction logic, and environmental responsibility as generative drivers of architectural form and spatial organisation.

A key objective is to explore how architecture can emerge through reciprocal relationships between material reuse, structural reasoning, and computational design. Students will examine how reclaimed materials can be digitally characterised, reorganised, and assembled through workflows combining material scanning, parametric modelling, structural optimisation, custom connector development, and mixed-reality-assisted construction.

Rather than adapting recycled or reclaimed materials to predetermined forms, the studio encourages design processes

in which architectural systems evolve directly from the physical characteristics, dimensions, condition, and availability of material resources.

Students will critically engage with the capacity of digital technologies to mediate between material intelligence, fabrication, and construction. Through scaled prototyping and an international design-build collaboration, the studio seeks to cultivate a mode of architectural practice in which circularity, computation, and tectonic thinking operate as an integrated framework.

Studio Project

1. Project Description

The primary studio project explores the design of a pavilion-scale architectural intervention that investigates circular tectonics through reclaimed materials, computational design, and digital fabrication. Rather than responding to a predetermined site or conventional building typology, each student will develop an autonomous architectural proposal within a notional full-scale spatial envelope of 3 m × 3 m × 3 m. This framework enables focused experimentation with structural systems, material organisation, spatial experience, fabrication processes, and assembly strategies.

Framed through the lens of Circular Tectonics, the project positions reclaimed materials as active generators of architectural form and construction logic. Students are required to develop their proposals through iterative computational workflows integrating material cataloguing, 3D scanning, parametric modelling, structural optimisation, digital fabrication, and mixed-reality-assisted assembly.

Design development will be continuously informed by the characteristics and availability of reclaimed materials, encouraging students to establish reciprocal relationships between digital design processes and physical material systems. The studio adopts a research-through-making approach. Each student will develop an individual architectural proposal and communicate it through drawings, computational models, fabrication documentation, and a scaled physical prototype produced through methods such as 3D printing, CNC machining, and laser cutting.

Building upon these individual investigations, students will collectively contribute to the design, fabrication, and assembly of a 1:1 installation during the Hong Kong intensive workshop. This shared design-build exercise will provide direct experience of collaborative fabrication, material reuse, assembly planning, and mixed-reality-assisted construction, while demonstrating how individual investigations can inform a collective architectural outcome.

2. Semester Organisation

Phase 1 – Circular Thinking & Material Exploration (Weeks 2–3)

Formal studio teaching commences in Semester Week 2. Students are introduced to the principles of Circular Tectonics through precedent studies, material investigations, and introductory computational design workflows.

This phase establishes a shared understanding of circular material systems, reclaimed construction components, material intelligence, and emerging digital fabrication technologies. Early exercises focus on precedent analysis, material documentation, digital cataloguing, introductory modelling, and small-scale tectonic experimentation.

Activities

- Introduction to Circular Tectonics and circular design principles
- Research into precedent projects and reclaimed material systems
- Material sourcing, classification, and digital catalogue development
- Introduction to Rhino, Grasshopper, Ameba, 3D scanning, and Fologram workflows
- Initial design speculation and tectonic exploration

Milestone:**Concept Proposal Pin-Up (Week 3)**

Students present an initial design agenda, precedent research, material investigation, digital material catalogue, and an early tectonic concept.

Phase 2 – Computational Design & Prototype Development (Weeks 4-8)

Students develop their individual architectural proposals within the 3 m × 3 m × 3 m design envelope. Through iterative computational workflows, proposals evolve in response to reclaimed material characteristics, structural reasoning, fabrication constraints, spatial intentions, and assembly requirements.

Students explore material-informed form generation, structural organisation, connector systems, fabrication methods, and physical prototyping. Regular tutorials, technical workshops, peer discussions, and design reviews support the progressive development of coherent architectural and tectonic strategies.

Activities

- Learning software (week 4-6)
- Parametric design development
- Material-informed form generation
- Structural optimisation and tectonic exploration
- Connector and joint development
- Digital fabrication testing
- Physical model-making and prototype testing
- Development of architectural drawings and computational diagrams
- Preparation for the Mid-Semester Review

Milestone:**Mid-Semester Review – Semester Week 8, Wednesday 16 September**

Assessment focuses on conceptual clarity, material intelligence, computational thinking, tectonic reasoning, prototype development, and evidence of an iterative design process.

Phase 3 – Design Resolution and Fabrication Planning (Weeks 9-13)

Following the Mid-Semester Review, students refine their individual proposals and develop them towards resolved architectural, material, structural, and fabrication outcomes.

Students are expected to demonstrate how material research, computational modelling, structural reasoning, fabrication processes, connector development, and assembly sequencing operate as an integrated design system.

Alongside their individual projects, students contribute ideas, material studies, fabrication knowledge, and assembly strategies towards the collective installation to be developed during the Hong Kong intensive workshop.

Activities

- Design refinement and spatial development
- Structural and tectonic resolution
- Material and fabrication testing
- Connector and joint development
- Scaled model production
- Assembly sequence development
- Preparation of drawings and fabrication documentation
- Collaborative planning for the Hong Kong installation
- Preparation for travel and workshop participation

Milestone:

Pre-Travel Review – Semester Week 13, Wednesday 21 October

Students present:

- Fully developed individual architectural proposal
- Material and fabrication strategy
- Structural and tectonic logic
- Scaled physical model
- Connector and assembly strategy
- Assembly sequence
- Drawings and fabrication documentation
- Proposed contribution to the Hong Kong design-build workshop

Assessment evaluates design resolution, fabrication feasibility, material integration, computational workflow, constructability, and communication quality.

Phase 4 – Hong Kong Intensive Design-Build Workshop (Weeks 14-15)

The travelling component of the studio takes place at the School of Architecture, The Chinese University of Hong Kong. During Semester Weeks 14 and 15, RMIT and CUHK students collaborate through an intensive design-build workshop combining computational design workflows with hands-on material preparation, fabrication, and assembly.

Rather than constructing their individual proposals at full scale, students collectively participate in the development and construction of a shared 1:1 installation using locally sourced reclaimed materials, digital fabrication techniques, and mixed-reality-assisted assembly methods.

The workshop also includes academic exchange, architectural visits, and engagement with Hong Kong's cultural, material, and urban contexts.

Activities

- Collaborative design development
- Reclaimed-material assessment and preparation
- Digital fabrication and component production
- Connector and assembly development
- 1:1 installation fabrication and assembly
- Mixed-reality-assisted construction
- Hong Kong Trip - Academic visits and cultural engagement
- International collaboration with CUHK students and staff
- Documentation of the design-build process

Milestone:

Hong Kong Design-Build Completion – Semester Week 15

Final Presentation at CUHK and Folio Submission – Monday 2 November

Students present their complete individual architectural proposals alongside documentation of the collaborative design-build installation. They reflect on the relationships between computational design, circular material systems, fabrication, collaboration, and architectural construction.

Final folio submissions include students' individual/group weekly work from week 2-13 and final projects, which communicate the complete design process, computational workflow, material research, fabrication logic, tectonic outcomes, and each student's identifiable contribution to the final 1:1 installation.

CUHK Exhibition Opening – Tuesday 3 November
Return to Melbourne – Wednesday 4 November

Phase 5 – RMIT Exhibition, Final Documentation and Reflection (Late Week 15 & Week 16)

RMIT exhibition preparation and installation.

Milestones

RMIT Exhibition Setup – Friday 6 November, Semester Week 15

RMIT Exhibition Opening – Friday 13 November, Semester Week 16

3. Teaching Structure Alignment

The studio teaching sequence commences in Semester Week 2 and establishes explicit relationships between:

- Circular material research → Material classification → Design criteria
- Material scanning → Computational modelling → Material-informed form generation
- Structural reasoning → Connector development → Fabrication strategy
- Individual proposal → Scaled physical prototyping → Design resolution
- Digital workflow → Assembly planning → Mixed-reality-assisted construction
- Individual investigation → Collaborative 1:1 installation → Exhibition and reflection

Each phase builds upon the preceding investigations, reinforcing iterative feedback between material exploration, computational design, structural reasoning, fabrication, and architectural making.

4. Mapping of Learning Tasks

The sequence of tasks makes explicit connections between:

- Material research → Material catalogue → Computational modelling
- Computational modelling → Structural reasoning → Fabrication strategy
- Individual proposal → Prototype → Collective installation
- Hong Kong workshop → Reflection → Exhibition

Each phase builds upon the previous, reinforcing iterative feedback between analysis and formation.

5. Task Differentiation

Individual Tasks

- Circular material and precedent research
- Material catalogue development
- Individual 3 × 3 × 3 m architectural proposal
- Computational modelling and optimisation
- Scaled physical prototype
- Final documentation and portfolio

Collaborative Tasks

- Material sourcing and testing
- Peer-review workshops
- Collective design discussions
- Design-build workshop with CUHK
- Fabrication and assembly of the 1:1 installation
- Final exhibition preparation

Each student is responsible for the authorship and resolution of their individual architectural proposal while contributing to the collective design-build installation and exhibition.

6. Assessment Format

Mid-Semester Review (Week 8)

Students are expected to present:

- Circular design concept
- Material research and catalogue
- Computational design development
- Structural and tectonic strategy
- Initial fabrication workflow
- Physical prototype and design models

Assessment focuses on conceptual clarity, material intelligence, computational thinking, tectonic reasoning, and iterative design development.

Pre-Travel Review (Week 13)

Students must present:

- Fully developed individual design proposal
- Material and fabrication strategy
- Structural logic
- Scaled physical model
- Assembly sequence
- Design documentation supporting participation in the Hong Kong workshop

Assessment evaluates design resolution, fabrication feasibility, material integration, computational workflow, and communication quality.

Studio Conceptual Agenda

Circular Tectonics investigates how reclaimed materials, computational design, and emerging fabrication technologies can establish new tectonic logics for architecture. Rather than treating material reuse as a constraint or a post-design sustainability measure, the studio positions reclaimed materials as active generators of architectural form, structural organisation, and construction methodology.

Drawing upon contemporary research in circular design, digital fabrication, computational modelling, and design-build practice, the studio explores how material intelligence can inform architectural decision-making from conception through fabrication and assembly. Particular emphasis is placed on the reciprocal relationship between digital workflows and physical making, enabling students to engage architecture as an iterative process of testing, adaptation, and construction.

Key areas of investigation include circular material systems, computational design, digital fabrication, structural reasoning, mixed-reality assisted assembly, and collaborative design-build practice. Through individual design exploration and international collaboration with the School of Architecture at The Chinese University of Hong Kong (CUHK), students critically examine how architecture can respond to environmental challenges through innovative, measurable, and buildable tectonic systems.

The studio asks:

- How can reclaimed materials become generators of architectural form and tectonic logic?
- How can computational design mediate between material availability, fabrication, and construction?
- How might collaborative design-build processes redefine contemporary architectural practice within a circular economy?

Through these questions, the studio promotes an architectural approach that integrates circularity, computation, material intelligence, and collaborative making as inseparable components of contemporary practice.

Indicative References / Key References

Shigeru Ban, Assemble Studio, Rotor, Lacaton & Vassal, The Living, Matter Design, Philip F. Yuan, Shuaizhong Wang....

2. Studio Skilling (100 words)

Studio skills are introduced progressively through research, computational design, material experimentation, and collaborative making. Students develop the capacity to engage reclaimed materials as design generators while integrating computational workflows with fabrication and assembly.

These skills are cultivated through iterative design exploration, digital modelling, physical prototyping, and design-build activities. Students learn to catalogue materials, develop computational design strategies, evaluate structural systems, and communicate fabrication logic through both digital and physical media.

Mastery is demonstrated through the resolution of an individual architectural proposal, the quality of the scaled prototype, and meaningful contribution to the collaborative installation and final exhibition.

Design & Communication Skills

Students are assessed on their ability to develop a coherent architectural proposal informed by circular design principles, material intelligence, and computational thinking. Design assessment focuses on the relationship between reclaimed materials, tectonic systems, fabrication strategies, and spatial experience rather than formal expression alone.

Students are expected to demonstrate iterative design development, evidencing how computational modelling, material experimentation, and prototype testing inform architectural decision-making. Success is measured through the consistency between conceptual ambition, material strategy, fabrication logic, and the final architectural outcome.

Communication is evaluated through the clarity and precision of drawings, computational workflows, physical models, fabrication documentation, and visual narratives. Digital and physical representations should clearly communicate structural organisation, material systems, assembly logic, and the evolution of the design process.

Teaching Framework

The studio adopts a research-led and practice-based teaching framework that integrates computational design, material experimentation, and collaborative making. Learning is structured through progressive phases that move from circular material research and computational exploration towards prototype development, international design-build collaboration, and public exhibition.

Teaching is delivered through weekly lectures, design tutorials, technical workshops, peer discussions, invited expert seminars, and formal review milestones. Continuous feedback supports iterative design development, encouraging students to refine their proposals through material testing, computational analysis, and physical prototyping.

The international collaboration with the School of Architecture at The Chinese University of Hong Kong extends the studio beyond the traditional design environment, allowing students to experience architecture as a collective process of design, fabrication, and construction. This framework encourages students to develop confidence in addressing contemporary environmental challenges through circular design thinking, interdisciplinary collaboration, and emerging digital technologies.

3. Studio Semester Schedule

RMIT Master of Architecture Design Studio Semester Schedule, Semester 2 2026		
Semester Week	Date	Activity
<u>0</u>	<u>Tuesday 14 July</u>	<u>Studio Leaders to present at Ballot Presentation between 4:00 pm - 9.00pm</u>
<u>1</u>	<u>Wed 22 July</u>	<u>Tutorial – No class</u>
<u>2</u>	<u>Wed 29 July</u>	<u>Tutorial – First class</u>
<u>3</u>	<u>Wed 5 August</u>	<u>Tutorial – Second class</u>
<u>4</u>	<u>Wed 12 August</u>	<u>Tutorial – Third class</u>
<u>5</u>	<u>Wed 19 August</u>	<u>Tutorial – Fourth class</u>
<u>6</u>	<u>Wed 26 August</u>	<u>Tutorial – Fifth class</u>
Mid Semester Break 31 August – 6 September No lectures and no tutorials		
<u>7</u>	<u>Wed 9 September</u>	<u>Tutorial -Sixth class</u>
<u>8</u>	<u>Wed 16 September</u>	<u>Tutorial – Seventh class / Mid Semester Review</u>
<u>9</u>	<u>Wed 23 September</u>	<u>Tutorial - Eighth class</u>
<u>10</u>	<u>Wed 30 September</u>	<u>Tutorial – Ninth class</u>
<u>11</u>	<u>Wed 7 October</u>	<u>Tutorial – Tenth class</u>
<u>12</u>	<u>Wed 14 October</u>	<u>Tutorial – Eleventh class</u>
<u>13</u>	<u>Wed 21 October</u>	<u>Tutorial –Twelfth class & Pre-Travel Review</u>
<u>14</u>	<u>Sunday 25 October to Saturday 31 October</u>	<u>Study, Travel, Model Making at CUHK – Hong Kong</u>
<u>15</u>	<u>Sunday 1 November to Tuesday 3 November</u>	<u>Final Presentation at CUHK & Folio Submission on Monday 2 November</u> <u>CUHK Exhibition on Tuesday 3 November</u> <u>Return to Melbourne on Wednesday 4 November</u> <u>RMIT Exhibition Setup on Friday 6 November</u>
<u>16</u>	<u>Monday 9 November Tuesday 10 November</u>	<u>Studio Leaders to attend Grade Moderation (scheduled 30min session)</u>
	<u>Friday 13 November</u>	<u>RMIT Exhibition Opening from 6.00pm</u>

4. Studio Assessment Framework

Design Skills Criteria

In this studio, students will be expected to:

- Demonstrate the ability to develop architectural proposals informed by circular design principles, integrating reclaimed materials, structural reasoning, computational design, and fabrication strategies within a coherent tectonic system.
- Develop an iterative design process that combines material exploration, computational modelling, physical prototyping, and critical reflection to inform architectural decision-making.

- Engage critically with reclaimed materials as active generators of architectural form, construction logic, and spatial organisation rather than as secondary sustainability considerations.
- Translate conceptual ideas into buildable tectonic systems through digital fabrication workflows, structural articulation, and assembly strategies.
- Position their work within contemporary architectural discourse by demonstrating awareness of circular economy principles, material innovation, collaborative design-build practice, and environmental responsibility.

In this Studio, students will be assessed on:

- The clarity and coherence of their circular design strategy, particularly the integration of reclaimed materials, structural systems, and computational workflows.
- The depth of iterative design exploration and the quality of material experimentation, prototype development, and computational investigation.
- Their ability to synthesise material intelligence, fabrication logic, and tectonic thinking into a resolved architectural proposal.
- The degree to which the final proposal demonstrates innovation, constructability, and meaningful engagement with circular design principles.

Communication Skills Criteria:

In this studio, students will be expected to:

- Use drawings, diagrams, digital workflows, and physical models to communicate architectural ideas, material systems, fabrication strategies, and tectonic relationships.
- Employ scaled physical prototypes and computational models to demonstrate structural organisation, material behaviour, and assembly logic.
- Integrate material studies, fabrication documentation, and process diagrams into presentations to communicate the evolution of design thinking.

In this studio, communication of design ideas will be assessed based on:

- The clarity and precision of drawings, models, and diagrams in communicating tectonic systems and material organisation.
- The effectiveness of physical prototypes in demonstrating fabrication logic, assembly sequence, and spatial qualities.
- The integration of computational workflows, material research, and fabrication processes within the overall design narrative.

Studio Progress:

In this studio, students are expected to demonstrate continuous development through iterative design exploration, material experimentation, computational modelling, physical prototyping, and collaborative learning. Formal studio teaching commences in Semester Week 2, and progress is monitored through structured milestones across the remainder of the semester.

Studio milestone activities and tasks include:

- Introduction to circular design, precedent research, and material investigation — Semester Weeks 2–3
- Concept Proposal Pin-Up — Semester Week 3
- Computational design development and prototype exploration — Semester Weeks 4–8
- Mid-Semester Review — Semester Week 8
- Design refinement, fabrication planning, and scaled model production — Semester Weeks 9–13
- Pre-Travel Review — Semester Week 13
- Hong Kong collaborative design-build workshop — Semester Weeks 14–15
- Final Presentation at CUHK, Folio Submission and CUHK Exhibition — Semester Week 15
- RMIT exhibition preparation — Semester Weeks 15–16
- RMIT Exhibition Opening — Semester Week 16

Students are expected to demonstrate increasing levels of design maturity, material intelligence, computational integration, fabrication understanding, and communication quality throughout the semester

Core Learning Outcomes – Master of Architecture

Upon successful completion of this course students will be able to:

1. Apply design knowledge to an architectural problem.
2. Critically analyse, evaluate and make informed judgement on a selected range of technical, programmatic, theoretical, historical and professional issues and their implications for the building design, as outlined by each studio program.
3. Integrate and make manifest a body of practical and theoretical knowledge into the design process, within the time deadlines given by each studio.
4. Investigate & comprehend architecture and its relationship towards location, program, form and representation.
5. Demonstrate & articulate design skills through an iterative and considered design process, to resolve ideas from concept formation through to design development.
6. Communicate design ideas to high standard, demonstrating through the building design & its representation the aims & claims that are made for the building design.

5. Assessment Grades

- **High Distinction (80-100%)** Exceptionally clear understanding of subject matter and appreciation of issues; well organised, with formulated and sustained presentation and response to critique. Addresses all the specific objectives with many to a high standard. Evidence of creative insight and originality.
- **Distinction (70-79%)** Strong grasp of subject matter and appreciation of key issues; addresses all the specific objectives, with several to a high standard; clearly developed presentation and response to critique. Evidence of creative and solid work.
- **Credit (60-69%)** Competent understanding of subject matter and appreciation of the main issues; addresses all the specific objectives, some reasonably well. Clearly developed presentation and response to critique; well prepared and presented.
- **Pass (50-59%)** Satisfactory. Appreciation of subject matter and issues. Addresses all the specific objectives; work generally lacking in depth and breadth. Often work of this grade demonstrates only basic comprehension or competency. Work of this grade may be poorly prepared and presented. Investment of greater care and thought in organising and structuring work would be required to improve.
- **Fail (0-49%)** Unsatisfactory. Evidence of lack of understanding of subject, minimal or inadequate comprehension and does not address all the objectives. Work is often inadequate in depth and breadth and sometimes incomplete or irrelevant; lack of care and thought in organising and structuring work.